

must be entered manually into the configuration settings of the internet connection of the client computer for it to be able to access the internet. Some ISPs have their own DNS Name Servers, else you can always use servers provided by OpenDNS or the Google Public DNS servers. For your own website, if you purchase your domain name and hosting from the same place, the DNS settings and registration will be taken care of automatically for you. However, if you bought a domain name and want to use free hosting provided by someone else, you will need to manually specify the Name Servers of your host.

Web pages and hyperlinks (HTTP)

If there was a popularity contest for protocols, the HyperText Transfer Protocol would surely be one of the top contenders. Consider the process to open a website:

You enter the URL of the website you want to visit, because you only know the domain name. The DNS Name Server parses the domain name and maps it to the corresponding IP address. Your request for the website reaches the Web Server, and it serves you back the page you requested. Of course, this



Isn't it just everywhere!

is one of the simplest case scenarios, but almost any interaction of a user and a website is handled according to [HTTP](#). That is why almost all URLs start with “[http://](#)” or “[https://](#)”. [HTTPS](#) is HTTP using Secure Socket Layer (SSL) for encryption.

HTTP was designed as a ‘request-response’ protocol, meaning that is modelled after how humans communicate in the real world. In general, a client sends a request to a server, which appropriately sends back a response. The response can also contain resources, if requested. The latest version of HTTP is HTTP 2.0 which was standardized in 2015. Prior to that we had HTTP 1.1, which is very much in use even today, because older technologies take time to phase out. HTTP (along with HTML) was invented by Tim Berners-Lee and his team at CERN, the original development starting in 1989.